MAR-ACC-45: Arista Access Node Hardware/Environmental Fault SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_3378

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss) on Arista Access nodes, specifically MAR-ACC-45, located at the Marathahalli Core site. The Access role is critical in providing connectivity to the network edge, and its failure can cause service disruptions. The underlying protocol mechanics are based on the Arista EOS (Extensible Operating System) which utilizes a proprietary ASIC (Application-Specific Integrated Circuit) to handle packet forwarding and switching operations. The ASIC memory parity errors and line card crashes can be attributed to hardware malfunctions or environmental factors such as power supply fluctuations or overheating. The loss of power supply redundancy can also lead to node failures. The Arista CLI dialect is used to troubleshoot and resolve these issues, and the expected output is based on the EOS platform.

Pre-requisite Checks

- pre-requisite: Verify the site location and node ID to ensure accuracy
- pre-requisite: Check the redundant link status to ensure connectivity
- pre-requisite: Dispatch remote hands to the Marathahalli Core site, if necessary

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal error, Expected Output: Error information for the system, Description: This command displays error information for the system, including ASIC memory parity errors and line card crashes.

Command: show power supply, Expected Output: Power supply status and redundancy information, Description: This command displays power supply status and redundancy information, which can help identify any issues with the power supply redundancy.

Command: show system internal crashinfo, Expected Output: Crash information for the system, Description: This command displays crash information for the system, including line card crashes and other
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```
relevant data.
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Resolution Steps

Step 1: Step: Power cycle the node, Command: reload, Description: This step involves restarting the node to clear any hardware-related issues or environmental factors.

Step 2: Step: Replace the faulty line card, Command: no module , Description: This step involves replacing the faulty line card with a new one to resolve line card crashes.

Step 3: Step: Check and replace the power supply, Command: show power supply, Description: This step involves checking the power supply status and replacing it if necessary to resolve power supply redundancy loss.

Post-Check Verification

```
Command: show system internal error, Expected Output: Error
information for the system, Description: This command displays error
information for the system to verify if the issue has been resolved.

Command: show power supply, Expected Output: Power supply status and
redundancy information, Description: This command displays power
supply status and redundancy information to verify if the power supply
redundancy issue has been resolved.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy run start, Description: This step involves restoring the original configuration to revert any changes made during the troubleshooting process.

- Step: Reboot the node, Command: reload, Description: This step involves rebooting the node to ensure that it returns to its previous state.